

4 Direct Proof

Use the method of direct proof to prove the following statements.

2. If x is an odd integer, then x^3 is odd.

Proof. Suppose x is an odd integer. Thus $x = 2a + 1$ for some $a \in \mathbb{Z}$. Consequently $x^3 = (2a + 1)^3 = 8a^3 + 12a^2 + 6a + 1 = 2(4a^3 + 6a^2 + 3a) + 1$. Thus $x^3 = 2b + 1$, where b is the integer $4a^3 + 6a^2 + 3a$. Therefore x^3 is odd by definition of an odd number. $\square$

4. Suppose $x, y \in \mathbb{Z}$. If x and y are odd, then xy is odd.

Proof. Suppose x and y are odd. Then $x = 2a + 1$ for some $a \in \mathbb{Z}$, and $y = 2b + 1$ for some $b \in \mathbb{Z}$. Consequently $xy = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$. Thus $xy = 2c + 1$, where c is the integer $2ab + a + b$. Therefore xy is odd by definition of an odd number. $\square$

6. Suppose $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Proof. Suppose $a \mid b$ and $a \mid c$. Then $b = ak$ for some $k \in \mathbb{Z}$ and $c = ah$ for some $h \in \mathbb{Z}$. Consequently $b + c = ak + ah = a(k + h)$. Thus $b + c = am$, where m is the integer $k + h$. Therefore $a \mid (b + c)$ by definition of divisibility. $\square$

8. Suppose a is an integer. If $5 \mid 2a$, then $5 \mid a$.

Proof. Suppose $5 \mid 2a$. Then $2a = 5k$ for some $k \in \mathbb{Z}$. Observe that $a = 6a - 5a = 3(2a) - 5a$. Substituting $2a = 5k$ gives $a = 3(5k) - 5a = 15k - 5a = 5(3k - a)$. Thus $a = 5n$, where n is the integer $3k - a$. Therefore $5 \mid a$ by definition of divisibility. $\square$